

5 k -Center

Consider the following application. Given a set of cities, with intercity distances specified, pick k cities for locating warehouses in so as to minimize the maximum distance of a city from its closest warehouse. We will study this problem, called the k -center problem, and its weighted version, under the restriction that the edge costs satisfy the triangle inequality. Without this restriction, the k -center problem cannot be approximated within factor $\alpha(n)$, for any computable function $\alpha(n)$, assuming $\mathbf{P} \neq \mathbf{NP}$ (see Exercise 5.1).

We will introduce the algorithmic technique of *parametric pruning* for solving this problem. In Chapter 17 we will use this technique in a linear programming setting.

Problem 5.1 (Metric k -center) Let $G = (V, E)$ be a complete undirected graph with edge costs satisfying the triangle inequality, and k be a positive integer. For any set $S \subseteq V$ and vertex $v \in V$, define $\text{connect}(v, S)$ to be the cost of the cheapest edge from v to a vertex in S . The problem is to find a set $S \subseteq V$, with $|S| = k$, so as to minimize $\max_v \{\text{connect}(v, S)\}$.

5.1 Parametric pruning applied to metric k -center

If we know the cost of an optimal solution, we may be able to prune away irrelevant parts of the input and thereby simplify the search for a good solution. However, as stated in Chapter 1, computing the cost of an optimal solution is precisely the difficult core of $\mathbf{NP}$ -hard $\mathbf{NP}$ -optimization problems. The technique of parametric pruning gets around this difficulty as follows. A parameter t is chosen, which can be viewed as a “guess” on the cost of an optimal solution. For each value of t , the given instance I is pruned by removing parts that will not be used in any solution of cost $\leq t$. Denote the pruned instance by $I(t)$. The algorithm consists of two steps. In the first step, the family of instances $I(t)$ is used for computing a lower bound on OPT , say t^* . In the second step, a solution is found in instance $I(\alpha \cdot t^*)$, for a suitable choice of α .

A restatement of the k -center problem shows how parametric pruning applies naturally to it. Sort the edges of G in nondecreasing order of cost, i.e., $\text{cost}(e_1) \leq \text{cost}(e_2) \leq \dots \leq \text{cost}(e_m)$, and let $G_i = (V, E_i)$, where $E_i =$

$\{e_1, e_2, \dots, e_i\}$. A *dominating set* in an undirected graph $H = (U, F)$ is a subset $S \subseteq U$ such that every vertex in $U - S$ is adjacent to a vertex in S . Let $\text{dom}(H)$ denote the size of a minimum cardinality dominating set in H . Computing $\text{dom}(H)$ is **NP**-hard. The k -center problem is equivalent to finding the smallest index i such that G_i has a dominating set of size at most k , i.e., G_i contains k stars spanning all vertices, where a *star* is the graph $K_{1,p}$, with $p \geq 1$. If i^* is the smallest such index, then $\text{cost}(e_{i^*})$ is the cost of an optimal k -center. We will denote this by OPT . We will work with the family of graphs $G_1, \dots, G_m$.

Define the *square of graph H* to be the graph containing an edge (u, v) whenever H has a path of length at most two between u and v , $u \neq v$. We will denote it by H^2 . The following structural result gives a method for lower bounding OPT .

Lemma 5.2 *Given a graph H , let I be an independent set in H^2 . Then, $|I| \leq \text{dom}(H)$.*

Proof: Let D be a minimum dominating set in H . Then, H contains $|D|$ stars spanning all vertices. Since each of these stars will be a clique in H^2 , H^2 contains $|D|$ cliques spanning all vertices. Clearly, I can pick at most one vertex from each clique, and the lemma follows. $\square$

The k -center algorithm is:

Algorithm 5.3 (Metric k -center)

1. Construct $G_1^2, G_2^2, \dots, G_m^2$.
2. Compute a maximal independent set, M_i , in each graph G_i^2 .
3. Find the smallest index i such that $|M_i| \leq k$, say j .
4. Return M_j .

The lower bound on which this algorithm is based is:

Lemma 5.4 *For j as defined in the algorithm, $\text{cost}(e_j) \leq \text{OPT}$.*

Proof: For every $i < j$ we have that $|M_i| > k$. Now, by Lemma 5.2, $\text{dom}(G_i) > k$, and so $i^* > i$. Hence, $j \leq i^*$. $\square$

Theorem 5.5 *Algorithm 5.3 achieves an approximation factor of 2 for the metric k -center problem.*

Proof: The key observation is that a maximal independent set, I , in a graph is also a dominating set (for, if some vertex v is not dominated by I , then $I \cup \{v\}$ must also be an independent set, contradicting I 's maximality). Thus,